

Triangle Graph Analysis

Feynkit Analysis

November 7, 2025

Abstract

This document presents a comprehensive analysis of a 1-loop Feynman integral with 3 internal propagators and 3 external legs. We compute the Symanzik polynomials, parametric representations, GKZ hypergeometric system, and toric ideal structure. All calculations were performed using Feynkit, a symbolic computation toolkit for Feynman integrals.

Contents

1	Introduction	2
2	Graph Properties	2
2.1	Topological Invariants	2
2.2	Edge Structure	2
3	Symanzik Polynomials	4
3.1	First Symanzik Polynomial	4
3.2	Second Symanzik Polynomial	4
4	Parametric Representations	5
4.1	Schwinger Parametrisation	5
4.2	Feynman Parametrisation	5
4.3	Lee-Pomeransky Parametrisation	6
5	GKZ Hypergeometric System	7
5.1	GKZ A -Hypergeometric System	7
6	Algebraic Structure	9
6.1	Toric Ideal and Integration-by-Parts Relations	9
7	Conclusion	9
A	Computational Details	9

1 Introduction

This report analyses the structure of a Feynman integral using modern algebraic geometry and differential equations techniques. We employ the parametric representation framework and study the integral through its associated GKZ A -hypergeometric system.

2 Graph Properties

We begin by describing the topological structure of the Feynman diagram.

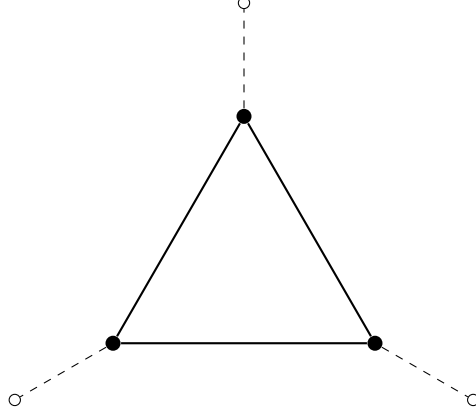


Figure 1: Feynman diagram topology. Internal vertices are shown as filled circles, internal propagators as solid lines, and external legs as dashed lines.

2.1 Topological Invariants

The graph has the following topological properties, summarised in Table 1.

Property	Value
Internal vertices	3
External legs	3
Internal propagators	3
Loop number	1

Table 1: Feynman graph topological properties.

2.2 Edge Structure

The complete list of edges is given in Table 2.

Index	Type	From	To	Mass
1	Internal	1	2	m_1
2	Internal	2	3	m_2
3	Internal	3	1	m_3
4	External	1	4	—
5	External	2	5	—
6	External	3	6	—

Table 2: Complete edge list for the Feynman graph.

3 Symanzik Polynomials

The Symanzik polynomials U and F encode the topological and kinematic structure of the Feynman integral. These polynomials are fundamental to the parametric representation.

3.1 First Symanzik Polynomial

The first Symanzik polynomial U depends only on the graph topology:

$$U = a_1 + a_2 + a_3 \quad (1)$$

This polynomial is homogeneous in the Schwinger parameters and encodes the sum over all spanning trees of the graph.

3.2 Second Symanzik Polynomial

The second Symanzik polynomial F includes kinematic information:

$$F = \frac{1}{\mu^2} \cdot \left(a_1^2 \cdot m_1^2 + a_1 \cdot a_2 \cdot m_1^2 + a_1 \cdot a_2 \cdot m_2^2 + \frac{s_{12}}{2} \cdot a_1 \cdot a_2 + \frac{s_{23}}{2} \cdot a_1 \cdot a_2 + a_1 \cdot a_3 \cdot m_1^2 + a_1 \cdot a_3 \cdot m_3^2 + \frac{s_{12}}{2} \cdot a_1 \cdot a_3 + \frac{s_{13}}{2} \cdot a_1 \cdot a_3 + a_2^2 \cdot m_2^2 + a_2 \cdot a_3 \cdot m_2^2 + a_2 \cdot a_3 \cdot m_3^2 + \frac{s_{13}}{2} \cdot a_2 \cdot a_3 + \frac{s_{23}}{2} \cdot a_2 \cdot a_3 + a_3^2 \cdot m_3^2 \right) \quad (2)$$

This polynomial encodes the momentum flow and mass structure of the diagram.

4 Parametric Representations

We present three standard parametric representations: Schwinger, Feynman, and Lee-Pomeransky. Each representation has distinct advantages for different computational tasks.

4.1 Schwinger Parametrisation

Schwinger parametrisation with 3 alpha parameters. Each $\alpha \in [0, \infty)$. Integration domain is $\mathbb{R}_+^3$.

Parameters. The integration parameters are:

$$\alpha_1, \quad \alpha_2, \quad \alpha_3 \quad (3)$$

Prefactor. The overall prefactor is:

$$\frac{e^{\epsilon \cdot \gamma_E}}{\Gamma(\nu_1) \cdot \Gamma(\nu_2) \cdot \Gamma(\nu_3)} \quad (4)$$

Measure. The integration measure is:

$$\alpha_1^{\nu_1-1} \cdot \alpha_2^{\nu_2-1} \cdot \alpha_3^{\nu_3-1} \quad (5)$$

Integrand. The integrand is:

$$(\alpha_1 + \alpha_2 + \alpha_3)^{-\frac{D}{2}} \cdot e^{\frac{1}{2 \cdot \mu^2 \cdot (\alpha_1 + \alpha_2 + \alpha_3)} \cdot (-2 \cdot \alpha_1^2 \cdot m_1^2 - 2 \cdot \alpha_1 \cdot \alpha_2 \cdot m_1^2 - 2 \cdot \alpha_1 \cdot \alpha_2 \cdot m_2^2 - \alpha_1 \cdot \alpha_2 \cdot s_{12} - \alpha_1 \cdot \alpha_2 \cdot s_{23} - 2 \cdot \alpha_1 \cdot \alpha_3 \cdot m_1^2 - 2 \cdot \alpha_1 \cdot \alpha_3 \cdot m_3^2 - \alpha_1 \cdot \alpha_3 \cdot s_{12} - \alpha_1 \cdot \alpha_3 \cdot s_{13} - 2 \cdot \alpha_2^2 \cdot m_2^2 - 2 \cdot \alpha_2 \cdot \alpha_3 \cdot m_2^2 - 2 \cdot \alpha_2 \cdot \alpha_3 \cdot s_{23} - \alpha_2 \cdot \alpha_3 \cdot s_{12} - \alpha_2 \cdot \alpha_3 \cdot s_{13})} \quad (6)$$

4.2 Feynman Parametrisation

Feynman parametrisation with 3 parameters on the unit simplex. Constraint: $\sum_i a_i = 1$, all $a_i \geq 0$.

Parameters. The integration parameters are:

$$a_1, \quad a_2, \quad a_3 \tag{7}$$

Prefactor. The overall prefactor is:

$$\frac{e^{\epsilon \cdot \gamma_E}}{\Gamma(\nu_1) \cdot \Gamma(\nu_2) \cdot \Gamma(\nu_3)} \cdot \Gamma\left(-\frac{D}{2} + \nu_1 + \nu_2 + \nu_3\right) \quad (8)$$

Measure. The integration measure is:

$$a_1^{\nu_1-1} \cdot a_2^{\nu_2-1} \cdot a_3^{\nu_3-1} \quad (9)$$

Integrand. The integrand is:

$$\begin{aligned} & (2 \cdot \mu^2)^{-\frac{D}{2} + \nu_1 + \nu_2 + \nu_3} \cdot (a_1 + a_2 + a_3)^{-D + \nu_1 + \nu_2 + \nu_3} \cdot (2 \cdot a_1^2 \cdot m_1^2 + 2 \cdot a_1 \cdot a_2 \cdot m_1^2 + 2 \cdot a_1 \cdot a_2 \\ & \cdot m_2^2 + a_1 \cdot a_2 \cdot s_{12} + a_1 \cdot a_2 \cdot s_{23} + 2 \cdot a_1 \cdot a_3 \cdot m_1^2 + 2 \cdot a_1 \cdot a_3 \cdot m_3^2 + a_1 \cdot a_3 \cdot s_{12} + a_1 \cdot a_3 \cdot s_{13} + 2 \\ & \cdot a_2^2 \cdot m_2^2 + 2 \cdot a_2 \cdot a_3 \cdot m_2^2 + 2 \cdot a_2 \cdot a_3 \cdot m_3^2 + a_2 \cdot a_3 \cdot s_{13} + a_2 \cdot a_3 \cdot s_{23} + 2 \cdot a_3^2 \cdot m_3^2)^{\frac{D}{2} - \nu_1 - \nu_2 - \nu_3} \end{aligned} \quad (10)$$

Constraints. The parameters satisfy the following constraints:

$$a_1 + a_2 + a_3 = 1 \quad (11)$$

4.3 Lee-Pomeransky Parametrisation

Lee-Pomeransky parametrisation with 3 u parameters. Each $u \in [0, \infty)$. Uses combined polynomial $G = U + F$. Integration domain is $\mathbb{R}_+^3$.

Parameters. The integration parameters are:

$$u_1, \quad u_2, \quad u_3 \quad (12)$$

Prefactor. The overall prefactor is:

$$\frac{e^{\epsilon \cdot \gamma_E} \cdot \Gamma\left(\frac{D}{2}\right)}{\Gamma(\nu_1) \cdot \Gamma(\nu_2) \cdot \Gamma(\nu_3) \cdot \Gamma(D - \nu_1 - \nu_2 - \nu_3)} \quad (13)$$

Measure. The integration measure is:

$$u_1^{\nu_1-1} \cdot u_2^{\nu_2-1} \cdot u_3^{\nu_3-1} \quad (14)$$

Integrand. The integrand is:

$$\begin{aligned} & \left(\sqrt{2} \cdot \mu\right)^D \cdot \left(2 \cdot m_1^2 \cdot u_1^2 + 2 \cdot m_1^2 \cdot u_1 \cdot u_2 + 2 \cdot m_1^2 \cdot u_1 \cdot u_3 + 2 \cdot m_2^2 \cdot u_1 \cdot u_2 + 2 \cdot m_2^2 \cdot u_2^2 \right. \\ & + 2 \cdot m_2^2 \cdot u_2 \cdot u_3 + 2 \cdot m_3^2 \cdot u_1 \cdot u_3 + 2 \cdot m_3^2 \cdot u_2 \cdot u_3 + 2 \cdot m_3^2 \cdot u_3^2 + 2 \cdot \mu^2 \cdot (u_1 + u_2 + u_3) \\ & \left. + s_{12} \cdot u_1 \cdot u_2 + s_{12} \cdot u_1 \cdot u_3 + s_{13} \cdot u_1 \cdot u_3 + s_{13} \cdot u_2 \cdot u_3 + s_{23} \cdot u_1 \cdot u_2 + s_{23} \cdot u_2 \cdot u_3\right)^{-\frac{D}{2}} \end{aligned} \quad (15)$$

5 GKZ Hypergeometric System

The Feynman integral satisfies a system of partial differential equations known as the GKZ (Gelfand-Kapranov-Zelevinsky) A -hypergeometric system. This system provides a systematic approach to studying the analytic properties of the integral.

5.1 GKZ A -Hypergeometric System

The GKZ hypergeometric system has 9 variables, 4 differential equations, and A -matrix of dimension 4×9 .

A -matrix. The A -matrix encoding the monomial exponent structure is:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 2 & 1 \end{bmatrix} \quad (16)$$

Newton polytope. The Newton polytope is the convex hull of the monomial support. It encodes the combinatorial structure of the polynomial $G(u) = U(u) + F(u)$. The polytope is visualised in Figure 2.

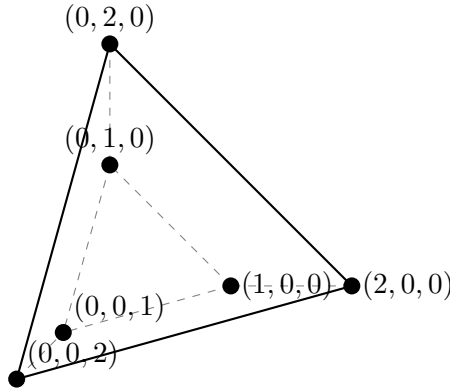


Figure 2: Newton polytope of the Lee-Pomeransky polynomial $G(u) = U(u) + F(u)$. Vertices represent monomial exponent vectors, solid edges are visible from the viewing angle, and dashed edges are hidden behind faces.

Parameter vector. The parameter vector β is:

$$\beta = \begin{pmatrix} -\frac{D}{2} + \nu_1 + \nu_2 + \nu_3 \\ \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} \quad (17)$$

Euler equations. The system consists of 4 Euler-type differential equations. These are Horn-type hypergeometric differential equations, where $\Phi = \Phi(z_1, \dots, z_9)$ represents the Feynman integral. Each equation has the form:

$$\begin{aligned} & z_1 \cdot \frac{\partial}{\partial z_1} \Phi + z_2 \cdot \frac{\partial}{\partial z_2} \Phi + z_3 \cdot \frac{\partial}{\partial z_3} \Phi + z_4 \cdot \frac{\partial}{\partial z_4} \Phi + z_5 \cdot \frac{\partial}{\partial z_5} \Phi + z_6 \cdot \frac{\partial}{\partial z_6} \Phi \\ & + z_7 \cdot \frac{\partial}{\partial z_7} \Phi + z_8 \cdot \frac{\partial}{\partial z_8} \Phi + z_9 \cdot \frac{\partial}{\partial z_9} \Phi = \left(-\frac{D}{2} + \nu_1 + \nu_2 + \nu_3 \right) \cdot \Phi \end{aligned} \quad (18)$$

$$2 \cdot z_1 \cdot \frac{\partial}{\partial z_1} \Phi + z_2 \cdot \frac{\partial}{\partial z_2} \Phi + z_3 \cdot \frac{\partial}{\partial z_3} \Phi + z_4 \cdot \frac{\partial}{\partial z_4} \Phi = \nu_1 \cdot \Phi \quad (19)$$

$$z_2 \cdot \frac{\partial}{\partial z_2} \Phi + 2 \cdot z_5 \cdot \frac{\partial}{\partial z_5} \Phi + z_6 \cdot \frac{\partial}{\partial z_6} \Phi + z_7 \cdot \frac{\partial}{\partial z_7} \Phi = \nu_2 \cdot \Phi \quad (20)$$

$$z_3 \cdot \frac{\partial}{\partial z_3} \Phi + z_6 \cdot \frac{\partial}{\partial z_6} \Phi + 2 \cdot z_8 \cdot \frac{\partial}{\partial z_8} \Phi + z_9 \cdot \frac{\partial}{\partial z_9} \Phi = \nu_3 \cdot \Phi \quad (21)$$

Each equation expresses a homogeneity property of the Feynman integral under rescalings of the Lee-Pomeransky parameters.

6 Algebraic Structure

6.1 Toric Ideal and Integration-by-Parts Relations

The toric ideal I_A is generated by 14 polynomials. These generators encode integration-by-parts (IBP) relations among the monomial coefficients of the Lee-Pomeransky polynomial.

Generators. The toric ideal is generated by:

$$z_1 \cdot z_5 - z_2^2 = 0 \tag{22}$$

$$z_1 \cdot z_6 - z_2 \cdot z_3 = 0 \tag{23}$$

$$z_1 \cdot z_7 - z_2 \cdot z_4 = 0 \tag{24}$$

$$z_1 \cdot z_8 - z_3^2 = 0 \tag{25}$$

$$z_1 \cdot z_9 - z_3 \cdot z_4 = 0 \tag{26}$$

$$z_2 \cdot z_6 - z_3 \cdot z_5 = 0 \tag{27}$$

$$z_2 \cdot z_7 - z_4 \cdot z_5 = 0 \tag{28}$$

$$z_2 \cdot z_8 - z_3 \cdot z_6 = 0 \tag{29}$$

$$z_2 \cdot z_9 - z_4 \cdot z_6 = 0 \tag{30}$$

$$z_3 \cdot z_7 - z_4 \cdot z_6 = 0 \tag{31}$$

$$z_3 \cdot z_9 - z_4 \cdot z_8 = 0 \tag{32}$$

$$z_5 \cdot z_8 - z_6^2 = 0 \tag{33}$$

$$z_5 \cdot z_9 - z_6 \cdot z_7 = 0 \tag{34}$$

$$z_6 \cdot z_9 - z_7 \cdot z_8 = 0 \tag{35}$$

Each generator represents a syzygy (polynomial relation) among the monomials, which corresponds to an IBP reduction identity for the Feynman integral. These relations can be used to express the integral in terms of simpler master integrals.

7 Conclusion

This analysis provides a complete mathematical characterisation of the Feynman integral. The parametric representations enable systematic evaluation, while the GKZ system and toric ideal structure reveal the underlying algebraic-geometric properties.

A Computational Details

All calculations in this document were performed using Feynkit, a Python package for symbolic Feynman integral computations. The source code is available at <https://github.com/feynkit/feynkit>.